

Chansoo (Charles) Kim

Graduate Researcher, Liquid-Metal Electronic Materials

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SUMMARY

Graduate researcher at Yonsei University (BLISS Lab, Prof. Jungmok Seo) developing sintering-free liquid-metal inks, wearable bioelectronics, and miniaturized electrochemical sensor toolkits. Bridging materials chemistry, embedded firmware, and biointerface engineering.

EDUCATION

2025 – present	M.S. student, Electrical & Electronic Engineering (Liquid-Metal Electronic Materials track) Yonsei University — BLISS Lab · Seoul, Republic of Korea - Advisor: Prof. Jungmok Seo (BLISS Lab) - Research focus: sintering-free liquid-metal inks, wearable bioelectronics, electrochemical biosensing
2019 – 2025	B.S., Electrical & Electronic Engineering Yonsei University · Seoul, Republic of Korea Bachelor of Science, College of Engineering

EXPERIENCE

2026-02 – 2026-06	Supervising Teaching Assistant — Capstone Design (전기전자종합설계) Yonsei University · School of Electrical & Electronic Engineering · Seoul, Republic of Korea Supervised an undergraduate capstone research team across a full design cycle, from problem definition through prototype and final demonstration. The team was selected as one of eight Outstanding Research Teams in the department.
2024-09 – 2025-12	Undergraduate Laboratory Instructor (Teaching Assistant) — Bio-electrical Electronics Lab Yonsei University · School of Electrical & Electronic Engineering · Seoul, Republic of Korea Led weekly laboratory sessions for the <i>Bio-electrical Electronics Laboratory</i> (바이오전기전자실험) course, translating research-grade biopotential / electrochemical / embedded instrumentation into hands-on exercises for undergraduate teams. Designed lab handouts, debugged student hardware and firmware, and mentored end-of-semester project work.
2020-04 – 2021-10	Training Operations Specialist — Education & Training Command (행정병) Republic of Korea Army · 35th Infantry Division Headquarters · Republic of Korea Active-duty military service supporting divisional education and training operations at the Headquarters' Education & Training Command. Coordinated training schedules, instructor assignments, and curriculum logistics across the division.
2020-08 – 2021-07	English Interpreter — U.S.–ROK Combined Joint Training Exercise U.S. Forces Korea ↔ Republic of Korea Army · Republic of Korea Selected as combined-exercise interpreter for two annual cycles (Aug 2020 and Jul 2021, ~1 month each), bridging communication between U.S. and ROK forces during joint operations.
2019-05 – 2019-12	Cultural Programs Coordinator — Cultural Planning Division (문화기획팀) Yonsei University · College of Engineering Student Council “PAGE” · Seoul, Republic of Korea Programmed and produced cultural events for the engineering student community as part of PAGE's freshman-year cultural planning division.

AWARDS & CERTIFICATIONS

2026-06	Outstanding Research Team Award — Capstone Design (전기전자종합설계) School of Electrical & Electronic Engineering, Yonsei University · Seoul RISE-Y Program Selected as one of eight outstanding capstone research teams, serving as the team's supervising teaching assistant. The award carries a teaching-assistant scholarship of KRW 1.25 million administered through the Seoul RISE-Y program.
2023-08	PTRD Scuba Diving Instructor Certification PTRD (2nd cohort, Scholarship Instructor Program) Earned the <i>Instructor</i> rating through PTRD's selective Scholarship Instructor Program (2nd cohort).

SERVICE & LEADERSHIP

2023-03 to 2023-08	President — Yonsei Skin-Scuba Diving Team (연세스킨스쿠버다이빙팀) Yonsei University Central Club Headed one of Yonsei's central student clubs; organized training dives, member certifications, and outreach programs for the campus diving community.
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RESEARCH AREAS

Liquid-Metal Materials

Catechol-Ga³⁺ chelation chemistry for sintering-free, primer-free EGaIn inks (PDA-LM). DFT (ORCA r2SCAN-3c) studies of amorphous Ga₂O₃ surface adhesion.

Wearable Bioelectronics

Stretchable / tattoo-sticker form factors integrating AD5941 (EC), ADS131M08 (EEG/EMG), and RHS2116 (neurostim). Apollo510 / DA14531 BLE.

Electrochemical Biosensors

AD5941-based 4-channel portable potentiostat (CV / amperometry / EIS / OCP). DIY SEBS-based screen-printed electrodes; Na⁺ ion-selective sensing.

Biointerface Coatings

PDRN + lubricant dual coatings for anti-fouling and tissue regeneration on suture and intramuscular EMG electrodes.

TECHNICAL SKILLS

Materials & Chemistry	EGaIn / liquid metal processing, Catechol-metal coordination, PDA / DOPA chemistry, Screen-printed electrodes (SPE), Lubricant / anti-fouling coatings
Computational	DFT (ORCA r2SCAN-3c, wB97X-D3/TZVP, BSSE), Python (data analysis, plotters), MATLAB
Embedded & Hardware	AD5941 (CV/CA/EIS/OCP), ESP32-C6, Apollo510, ADS131M08, RHS2116, KiCad PCB / sPCB / FPCB design, BLE (nRF52 / DA14531)
Characterization	UV-Vis, ITC, XPS interpretation, EIS, Cyclic voltammetry

LANGUAGES

Korean	Native
English	Professional working proficiency (served as interpreter during U.S.-ROK Combined Exercises)